Chemistry — MDCAT 2026 master notes

Every distinct Chemistry MCQ in the corpus, as one fact line. **1,266 fact buckets** built from **1,644 parsed questions** across 25 papers and mocks. `×n` marks how many corpus questions ask that fact — anything marked `×3` or more is a near-certain repeat. No explanations by design.

Facts are grouped by sub-topic inside each unit so the notes read as revision material; a few facts sit in a unit because that is where the paper put them, and are tagged `→uN` when they really belong elsewhere. Coverage is auditable via `../Analysis/notes_buckets.json`.

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u1 — Fundamental Concepts (moles, stoichiometry, yield, solutions)

The mole

- 1 mole = **6.02 × 10^{23}** particles (Avogadro's number). At STP one mole of gas = **22.4 dm^3**.
- **2 moles of water** = 3 atoms per molecule, so **1.92 × 10^{24} atoms** total. ×7
- Molecules in **9 g of ice** = 0.5 mol = **3.01 × 10^{23}**.
- Molecules in **22.4 dm^3 of H_2 at 0 °C, 1 atm** = **6.02 × 10^{23}** (1 mole).
- Moles of CO_2 containing **8.00 g of oxygen** = oxygen is 0.5 mol, so CO_2 = **0.25 mol**.
- Moles in an element = **mass / molar mass**, i.e. directly proportional to the **mass** of the element. ×2
- In `PV = nRT`, `n` = **N / N_a** (number of particles divided by Avogadro's number).
- Same moles of H_2, N_2 and O_2 in the same volume at STP → the **number of molecules are equal**.
- 10 g glucose in 100 cm^3 solution → **0.55 M**.
- **Molarity** = moles of solute per **dm^3 of solution** (not per dm^3 of water).
- **Mole fraction** = moles of one component over moles of **all components in a solution**.
- **Boiling point constants** are used to determine **molar masses** (ebullioscopy).
- Raoult's law: relative lowering of vapour pressure = **mole fraction of solute**.

Molar-mass arithmetic

- **60 g NaOH = 1.5 mol** (60 ÷ 40). The option list in the papers is damaged — work the division, not the letters.
- **0.1 g of sodium** = 0.1 ÷ 23 = **4.3 × 10^{-3} mol**. ×2
- **1.75 kg $CaCO_3$** = 1750 ÷ 100 = **17.5 mol**.
- **2.8 g ethylene (C_2H_4, M = 28)** = **0.1 mol**.
- **10.6 g Na_2CO_3** contains **0.3 mol of oxygen** (0.1 mol salt × 3).
- **16.5 g HCl** gives **0.452 mol NaCl** in the neutralisation.
- **3 mol calcium** = 3 × 40 = **120 g**.
- Oxygens in **278 g $FeSO_4 \cdot 7H_2O$** (M = 278) = 1 mol salt × 11 O = **6.62 × 10^{24} oxygen atoms**.

Balanced equations and limiting reagent

- Combustion of **2 mol propane** needs **10 mol O_2** ($C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$). ×6
- **$N_2 + 3H_2 \rightleftharpoons 2NH_3$**: 2.5 mol N_2 gives **5 mol NH_3**.
- **2A + B → product** with equal moles of A and B → **A is limiting**.

- For **2A + B_2 → 2AB**, A is limiting in the **2 mol A + 1 mol B_2** mixture.
- The reactant consumed **first** is the **limiting reactant** — the one **not in excess**. ×3
- 4 mol H_2 + 2 mol O_2 react exactly ($2H_2 + O_2 \rightarrow 2H_2O$: 4 mol H_2 needs 2 mol O_2) → **4 mol water**. ×2
- **$CS_2 + 3O_2 \rightarrow CO_2 + 2SO_2$**: 7.6 g CS_2 with 12.8 g O_2 → **O_2 is limiting**.

Yield

- **Percentage yield = actual ÷ theoretical × 100**. ×12
- Theoretical 25 g, actual 20 g → **80%**. ×9
- Expected 28 g, obtained 18 g, 4 g lost to spillage → real yield **14 g, 50%**. ×2
- % yield 80, actual 20 g → theoretical = **25 g**.
- Actual yield reaches the theoretical value at **100% (1.0)**.
- Reaction efficiency is checked with **percentage yield**, not actual or theoretical alone.
- Actual yield is **less than** theoretical because of **mechanical losses**, reaction **reversibility**, and not every molecule having activation energy — "**the substance is pure**" is the incorrect reason.

Empirical and molecular formula

- **Empirical formula** = simplest whole-number ratio of atoms. ×2
- Empirical formula of **glucose $C_6H_{12}O_6$** = **CH_2O**.
- Empirical **C_3H_3O** with molar mass 110.15 → multiply by 2 → **$C_6H_6O_2$**. ×2
- Polymer with empirical **CH_2** and molar mass **28000** → **2000 times** the empirical formula. ×2
- Combustion analysis: C = 40%, H = 6.67%, O = 53.3% → **CH_3O**... work the ratios: C:H:O = 3.33 : 6.67 : 3.33 → **CH_2O** as keyed by the papers.
- Phosphorus oxide with 43.6% oxygen → **P_2O_5**.
- Combustion of 0.5439 g giving 1.039 g CO_2 and 0.6369 g H_2O → **CH_4O**.

Isotopes and atomic mass

- Isotopes: **same atomic number, different mass number** — differ in **neutron** number. ×3
- Standard for relative atomic mass = **Carbon-12**. ×2
- Atomic masses depend on the **number of isotopes and their abundance**.
- **Mass spectrometry** gives the type and relative amount of each isotope; the detector measures **relative abundances**.
- **Boron** average mass 10.8 with isotopes 10 and 11 → **20%** of mass 10.
- Neon (^{20}Ne 90.92%, ^{21}Ne 8.82%, ^{22}Ne 0.26%) → **A_r = 20.18**; no single Ne atom is 20.18 u because that is the **average atomic mass**. ×2
- Isotopes are separated after converting the substance to a **charged state**.
- Same electrons and protons but different neutrons = **isotopes**.

Miscellaneous

- Stoichiometry is the **quantitative** study of reactants and products via a balanced equation.
- "**Active mass**" in the law of mass action = **moles per dm^3** (concentration).
- The equilibrium constant is **unitless when the moles of products equal the moles of reactants**.
- 'n' in the **Nernst equation** = number of electrons transferred in the redox reaction.

- Oxidation numbers X +6, Y −2, Z −1 combine as **XY_2Z_6**-type — the keyed option is **XY_2Z** (check charge balance: 6 + 2(−2) + n(−1) = 0 → n = 2).
- Air molecules do not settle because of the **elastic collisions** of gas molecules.
- RMS speed of a gas of N molecules = $\sqrt{(v_1^2 + v_2^2 + \ldots + v_n^2)/N}$.
- Gas density in a cube of side `a`, N molecules each of mass `m` = **Nm/a^3**.
- Two gases at the same T and P with $V_1:V_2$ = 2:3 → mole ratio **2:3**.
- **$2XeF_6 + SiO_2 \rightarrow 2XeOF_4 + SiF_4$**: 122.6 g XeF6 with 60 g SiO_2 — identify the limiting reagent by molar ratio.

Bled in from other units (same buckets)

- **In anisotropic crystals, refractive index is the property that does not change** with direction (electrical and thermal conductance do). →u5
- **In a mass spectrometer the collector measures relative abundances.** →u1 (stays)
- **Mole fraction** among CH_3OH (16 g), C_2H_5OH (92 g) and H_2O (36 g): CH_3OH 0.5, C_2H5OH 2.0, H_2O 2.0 → **mole fraction of C_2H_5OH and H_2O is the same**.
- **α decay** lowers Z by 2 and A by 4; **β^- decay** raises Z by 1, A unchanged. →u20
- U-232 with one α then one β gives **Z = 91** (90 then 91). →u20
- Two α particles emitted → **Z decreases by 4, A decreases by 8**. →u20
- X → Y + 3α → Y = **Z − 6, A − 12**. →u20
- W →(β) X →(α) Y →(α) Z: Z change from W to Z = **decreases by 3**. →u20
- β then α decay: **x = t + 4**. →u20
- Th-232 (Z 90) decaying by 6 α and 4 β → **Z = 82, A = 208**. →u20
- Chemistry debris: **PVA** is the resin binder for compact discs; English fragment "Kathie would have approved... and also have been amused".

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u2 — Atomic Structure (quantum numbers, configurations, spectra)

The four quantum numbers

- **n (principal)** → the shell, and it decides the orbital's **size**; s orbitals keep the same shape but grow as n rises. ×3
- **l (azimuthal)** → the subshell and the **shape** of the orbital. ×7
- **m (magnetic)** → the number of **orientations** in space.
- **s (spin)** → +½ or −½, the **direction of spin**; the one quantum number **not** from the Schrödinger equation.
- An electron is completely described by **four** quantum numbers.
- Maximum value of l is **l = n − 1** (l runs 0 → n−1). ×4
- **m ranges from −l to +l**: for l = 1, m = **−1 to 1**. ×3
- Orbitals in a subshell = **2l + 1**; for l = 2 (d) that is **5 orbitals** holding **10 electrons**. ×3
- **f (l = 3) has 7 orientations**; **p has 3**; **d has 5**; **s has 1** — the only single- orientation orbital. ×3
- Maximum electrons in shell n = **$2n^2$**.
- The **N shell (n = 4)** holds **s, p, d and f**.

Shapes and overlap

- **p orbitals** are dumb-bell shaped with **2 lobes**. ×2

- The π bond of **propene** is a sideward overlap of unhybridised **p orbitals**, not of the sp^2 hybrids. ×3
- In **sp^2** hybridisation the unhybridised p orbital is **perpendicular** to the plane of the three sp^2 orbitals — at **90°**. ×2
- **Benzene's** π molecular orbitals are **delocalised**.
- Ethene: **2 unhybridised p orbitals** overlap to form the π bond.
- Among $(CH_3)_3COH$ / CO_2 / C_2H_4 / CH_3COOH, the one where **every** carbon uses **sp^3** is **$(CH_3)_3COH$**.

Filling rules

- **Aufbau** = build up in order of increasing energy, ordered by the **(n + l) rule**.
- **Hund's rule** = orbitals of a subshell are singly occupied before any pairing. ×3
- **Pauli exclusion** = no two electrons share the same four quantum numbers. ×2
- After **3d** fills, the next electron enters **4p** (4s is already full). ×2
- After **6d** fills, the next electron enters **7s**. ×3
- Energy order: **4s < 3d < 4p**; and **4s < 4p < 4d < 4f**.

Configurations that repeat

- **[Ar] $4s^1$ $3d^5$** = chromium (Z 24). ×4
- **Fe^{2+}** = **[Ar] $4s^0$ $3d^6$** — the 4s electrons leave first.
- **Carbon** = $1s^2$ $2s^2$ $2p^2$. ×2
- **Argon** = $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ (18 e^-). ×3
- Outermost **ns^2 np^5** = **chlorine**; **ns^2np^6** = a halide ion / noble gas. ×2
- **Alkaline earth metals** = **ns^2**.
- **[Kr] $5s^2$ $4d^6$** = **ruthenium (Ru)**.
- **Au (Z 79)** = [Xe] $4f^{14}$ $5d^{10}$ **$6s^1$**.
- **$3d^6$ $4s^2$** in the 3d series = **nickel**.
- The two **anomalous** 3d configurations are **chromium and copper**, stabilised by half-filled / fully-filled d subshells. ×2
- The unpaired electron of a **chlorine atom** = **(3, 1, 1)**.
- **M^{2+}** with 2,8,14 and atomic weight 56 → Z = 26 (Fe), so **30 neutrons**.
- **Be > B in first ionisation energy** because Be has the **stable filled $2s^2$** configuration.
- Allowed set of quantum numbers: **n=3, l=2, m=0, s=+½** (l must be < n, |m| ≤ l).
- **Not allowed**: n=2, l=2 (l cannot equal n).
- The element with **$1s^2 2s^2 2p^1$** belongs to **group IIIA**.
- **$4s^2 4p^1$** valence shell = **group IIIA**.
- **[Ne] $3s^2 3p^5$** = a **non-metal of group VII-A**.
- Fully occupying the first two spherical (s) orbitals = **beryllium** ($1s^2 2s^2$).
- Pair with **1 electron in the outermost s**: **Na & Cr** (Na $3s^1$, Cr $4s^1$).

Magnetism

- **F_2 is diamagnetic** because all its electrons are **paired in p orbitals**. ×6
- Diamagnetic = no unpaired electrons; paramagnetic = at least one.
- Transition-metal compounds with unpaired electrons are **attracted by a magnet**.

Spectra of hydrogen

- Rydberg: $\bar{\nu} = R(1/n_1^2 - 1/n_2^2)$, $n_2 > n_1$. ×2
- **Balmer series** = the **visible** region. ×3
- **Lyman** = ultraviolet, and the **highest-frequency** series; **Paschen, Brackett, Pfund** = infrared. ×2
- Infrared transition = **n = 5 → 3** (3→2 and 4→2 are Balmer; 2→1 is Lyman).
- The last Lyman line ($n_2 \to \infty$) has **λ = 1/R**.
- $\bar{\nu} = R(1/2^2 - 1/n^2)$ with n > 2 is the **Balmer** equation.
- Orbital **energy** is decided by the **(n + l) rule**.

Historical particles and effects

- **J. J. Thomson** measured the **e/m ratio** of the electron.
- **Chadwick** discovered the **neutron**.
- **Cathode rays** are material particles with definite mass and velocity but they do **not** travel at the speed of light — that is the false statement.
- Cathode-ray nature is **independent of the gas** in the discharge tube.
- **Canal (positive) rays** = the other name for positive rays; proton mass **1.6726×10^{-27} kg**, charge/mass **9.54×10^7 C/kg**, deflected by electric and magnetic fields — the false claim is that they **do not cause mechanical motion**.
- Lightest canal-ray particles come from **H_2**.
- **Electric current in solution** is carried by **ions**, not free electrons. ×2
- **Zeeman effect** = splitting of spectral lines in a **magnetic** field; **Stark** = electric.
- **Bohr** introduced **energy levels** (stationary orbits).
- Same-orbital electrons spin opposite to **reduce repulsion** and cancel magnetic fields.
- Spontaneous electron emission from solids under radiation = the **photoelectric effect**.
- Of the three radioactive paths in an electric field, path 1 (bent to the negative plate) is **alpha**.
- **Planck's constant h is unchanged** when photon frequency doubles (E = hf scales with f).
- **Cu^{2+}** solution colour comes from **d → d** transitions.
- Racemic mixture in **S_n1** arises from the carbocation's **empty p orbital**.

Bled in from other units (same buckets)

- **Alkanes** have only **single** bonds between carbons. →u14
- **Fischer esterification** is a **condensation** reaction. →u18
- **760 torr = 101325 Pa**; **10 Pa = 0.000098 atm**. →u3

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u3 — Gases (gas laws, kinetic theory, real gases)

The laws

- **Boyle**: at constant T, **V ∝ 1/P** (PV = constant); double the pressure and volume becomes **one-half**. Sea-level air being dense is Boyle's law in practice. ×2
- **Charles**: at constant P, **V ∝ absolute temperature** — the law used to calculate **absolute temperature**. ×2
- **Avogadro**: equal volumes of gases at the same T and P hold equal numbers of molecules.

- **Dalton**: partial pressure $P_i = X_i P_t$ (mole fraction × total pressure). ×6
- The SI unit of pressure is the **pascal**.
- **$R = 0.0821$ atm dm^3 mol^{-1} K^{-1}**; **$R = 8.314$ J mol^{-1} K^{-1}**. ×3
- STP in the papers = **14.7 psi and 0 °C** (1 atm, 273 K).
- **1 atm = 760 torr = 101325 Pa**; **10 Pa = 0.000098 atm**.

Absolute zero and kinetic theory

- **Absolute zero = −273.16 °C** (= 0 K): molecular motion **ceases**, K.E. is **zero**, and an ideal gas's pressure is **0 atm**. ×3
- The postulates explaining ideal-gas behaviour = the **kinetic molecular theory of gases**.
- Ideal-gas molecules have **zero** intermolecular forces. ×6
- K.E. per molecule = **$3RT / 2N_a$**.
- Real gases deviate most at **low temperature and high pressure**. ×2
- Real gases show **positive** deviation as pressure rises because of **repulsive forces**.
- Both ideal and real gases have molecules in **constant random motion** (elastic collisions and zero size are ideal-only).
- The false KMT postulate: molecules "do not exert pressure when colliding with the wall" — they do.
- Density in a rigid sealed container **stays the same** when T rises.
- Compressibility factor **$Z = PV/RT > 1$** at STP → molar volume **greater than 22.4 L**.

Gas calculations that repeat

- Burning **500 dm^3 of ethene** at STP needs **1500 dm^3 of O_2** ($C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$).
- **$CaCO_3 \rightarrow CaO + CO_2$**: 48.8 L of CO_2 at STP (≈2.18 mol) needs about **200 g $CaCO_3$**... work moles × 100; the keyed answer is **200 g**.
- **$H_2 + ½O_2 \rightarrow H_2O$**: making **1 mol water** needs **11.2 dm^3 of H_2** at STP (½ mol).
- A gas at 10 dm^3 and 10 °C halves its volume at **141.5 K** (Charles: $T_2 = 283/2$).
- 1 atm N_2 at 25 °C raised to 2 atm at 50 °C: 1 dm^3 → **0.542 dm^3**.
- **X mL of H_2** effuses in 5 s; the same volume of **O_2** takes **20 s** (rate ∝ $1/\sqrt{M}$, O_2 is 16× denser → 4× slower).
- Boiling water at **110 °C** needs an external pressure of **760–1200 torr**.
- Jar at 830 mmHg with water drops, cooled 1% (vapour pressure 30 → 25 mmHg): new pressure **805 mmHg** — the keyed option is **792 mm of Hg**.
- **$KClO_3 \rightarrow KCl + O_2$**: the volume of solid KCl cannot be determined from gas data — answer "**cannot be determined from the given information**".
- Two ice cubes pressed into one = **H-bonding** (recrystallisation), not van der Waals.
- **Le Chatelier's principle**: a stress on an equilibrium is partly nullified by the system. ×2
- **Kc** for $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ has units **mol/dm^3**.

More gas facts

- One mole of any gas at STP occupies **22.414 dm^3**.
- **PV/nRT** is the **compressibility factor**.
- The general gas equation **PV = nRT** is the **ideal gas equation**.
- **R's dimensions** = **$[ML^2T^{-2}K^{-1}]$**; raising 1 mol of ideal gas by 1 K costs **8.314 J** (the papers key **0.0821 kJ**-scale options — use 8.314 J mol^{-1} K^{-1}).

- **Cp = Cv + R**, so with Cv = 3/2 R, **Cp = 5/2 R**; for a monatomic gas **Cp/Cv ≈ 1.67**.
- **Isothermal expansion**: temperature stays constant (internal energy of an ideal gas depends only on **temperature**).
- Absorbs 1000 J and expands against 20 kPa from 25 to 50 cm^3: ΔU = 1000 − 500 = **+500 J**.
- Work done expanding 15 → 20 dm^3 against 2 atm = **−5 atm dm^3** (W = −PΔV).
- Melting 1 mol ice at 0 °C absorbing 6025 J, ΔV = −0.002 dm^3: **ΔE ≈ 6025.20 J**.
- Bread + peanut butter (20 g CHO, 10 g protein, 9 g fat) = **201 kcal** total.
- **ΔH is measured indirectly by Hess's law**.
- Collecting gas **over water** is an application of **Dalton's law**.
- Stoichiometry must obey the **law of conservation of mass**.
- **RMS velocity ∝ 1/√(molar mass)**; v_rms = √(3P/ρ).
- H_2 vs O_2 at 300 K (O_2 16× heavier): H_2's rms speed is **4×** oxygen's.
- Boyle's law graph = **V vs 1/P, a straight line through the origin**.
- At absolute zero, hydrogen molecules have **all motion ceased**.
- Gas density depends on **pressure, temperature and molar mass** — all of the above.
- Partial pressure of H_2 in 9.6% NH_3 / 22.6% N_2 / 67.8% H_2 at 50 atm = **67.8 × 50/100**.
- Charles's law: gas volume would reach zero at **−273.15 °C**.
- **212 °F = 373 K**... 212 °F = 100 °C = **373 K**.
- Collisions must be **elastic** for total K.E. to stay constant.
- **Hydrogen bonding** stabilises α-helices and β-sheets of proteins and holds the two DNA strands together. ×2
- Mg + ½O_2 → MgO, ΔH = −692 kJ/mol: that is a standard enthalpy of **formation (ΔH°f)**.
- 125 g Mg + 32 g O_2 → MgO: **excess Mg left = 105 g** (O_2 is limiting; the keyed option list is damaged — work it from 2Mg + O_2).
- KMT explains **the behaviour of gases**.
- One corpus line here is an OCR fragment (a "Strain Stress / IDEAL GASES" heading) with no recoverable question.

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u4 — Liquids (intermolecular forces, vapour pressure, boiling point)

Intermolecular forces, strongest to weakest

- **Hydrogen bonding** is the strongest of the usual intermolecular set (London / dipole-dipole / H-bond — H-bond wins). ×2
- The **weakest** is the **instantaneous dipole–induced dipole (London dispersion)** force.
- Forces arising from **temporary dipoles** = **dispersion forces**.
- **H-bonding is strongest in HF**, but **H_2O boils higher** because water forms **more** hydrogen bonds. ×2
- **Alcohols boil higher than alkanes** of similar mass because of **hydrogen bonding**. ×6
- **Carboxylic acids** boil high because of **intermolecular hydrogen bonding** (dimerisation).
- **Ether boils lower than alcohols/phenols** — weaker intermolecular forces (no O–H to donate).
- **Diethyl ether's** low boiling point vs water = **weak intermolecular forces**.
- **Water > petrol** in boiling point because water's intermolecular forces are **stronger**.

- **H-bonding holds the two DNA strands together** and stabilises α-helix / β-sheet proteins.
- **H-bonding does NOT exist in SiH_4**; it does in proteins, alcohols and liquid HF.
- The polar part of soap dissolves in water via **dipole-dipole / ion-dipole** attraction.
- **I_2 molecules** attract by **instantaneous dipole–induced dipole** forces; iodine is the soft, weak-force molecular solid among H_2O / I_2 / sugar / graphite.
- **Molecular crystals** are held by **van der Waals forces**.
- **Sulphur melts higher than phosphorus** because S_8 has **stronger van der Waals forces**.
- **Group VIIA hydride with the lowest boiling point = HCl**. ×6
- Boiling-point order of the hydrides: **H_2O > HF > NH_3 > HCl**. ×2
- Alkali metals: **K has the least boiling point** of Li / K / Na / Cs.

Vapour pressure and boiling

- Vapour pressure does **not** depend on the **amount of liquid** (it depends on T, intermolecular forces and the type of bond). ×6
- Vapour pressure is **independent of the surface area** of the liquid.
- **Strong dipole-dipole forces → high heat of vaporisation**. ×2
- Heat absorbed when 1 mol of liquid becomes gas at its boiling point = **heat of vaporisation**. ×2
- Heat to turn 1 mol of solid **directly** into vapour = **standard heat of sublimation**; solid → gas directly = **sublimation**. ×2
- **Lowest vapour pressure at 20 °C = water** (strongest H-bonding of ether / chloroform / CCl_4 / water). ×2
- **Highest vapour pressure** of acetone / acetaldehyde / isopentane / benzene = **isopentane** (weakest forces, lowest mass).
- **Lower vapour pressure** of $CHCl_3$ / CCl_4 / H_2O / acetone = **water**. ×2
- If intermolecular forces between gas molecules suddenly vanished, **pressure would remain unchanged**. ×2
- **Vacuum distillation lowers the boiling point**: glycerin (290 °C at 1 atm) boils near **210 °C at 50 torr**.
- **Branching decreases boiling point** because it **reduces the surface area** available for attractive forces.
- Rate of evaporation is **not** affected by the **amount of liquid** (surface area, temperature and intermolecular forces do affect it).
- **Static equilibrium** is the equilibrium stage not affected by temperature. ×2
- Decomposition (vs evaporation) of 1 mol needs **high energy**.

Exothermic and endothermic

- **Oxidation of sulphur** is exothermic (melting ice, evaporating water and photosynthesis are endothermic).
- **Respiration** is the exothermic process among evaporation / sublimation / boiling / respiration.

Boiling point defined, and pressure effects

- Boiling point = the temperature at which **vapour pressure equals the atmospheric pressure**.
- Halving the external pressure makes ethanol boil **below 78 °C**.
- At **high altitude** water boils below 100 °C because of **lower atmospheric pressure**.
- **Steam burns worse than boiling water** because of its **high latent heat of vaporisation**.
- **Molar heat of vaporisation of water = 40.7 kJ/mol**.
- Highest boiling point of the alcohols = **n-propyl alcohol** (straight chain beats branched; tert-butyl is the most branched).

- Highest boiling point of the alkanes = **$C_{10}H_{22}$** (longest chain, strongest dispersion).
- **Lowest boiling point** of water / ethanol / H_2S / acetic acid = **hydrogen sulphide** (no H-bonding).
- **SO_2 has a higher heat of vaporisation than CO_2** because **SO_2 is polar and CO_2 is non-polar**.
- **HF's acidity is exceptionally low** because of its **strong H–F hydrogen bonding**.
- **Weakest London dispersion = He** (smallest, fewest electrons).
- **London dispersion is the only force among helium atoms** in the gas phase.
- **Hydrogen bonding is exhibited by NH_3** (not H_2S, HI or SiH_4).
- **Hydrocarbons do NOT form hydrogen bonds** (ammonia, ethyl alcohol and carboxylic acids do).
- **Highest surface tension = mercury** overall; among water / benzene / alcohols / ether it is **water**. ×2
- **Vapour pressure is measured with a manometer**.
- Vapour pressure does **not** depend on the **amount of liquid**.
- Dipole-dipole strength does **not** affect **moles** (it affects B.pt, heat of sublimation and heat of vaporisation).
- **Ice at 0 °C occupies more volume than water** because of the **empty spaces** in its open lattice; ice floats for the same reason. ×2
- **Freezing point** = the temperature where solid and liquid have the **same vapour pressure**.
- **Transition temperature** = where one crystalline form changes into another.
- 18 g glucose in 1 kg water boils at **100.52 °C** ($\Delta Tb = Kb \cdot m = 0.52 \times 1$).
- Benzene as solvent has the **largest molal boiling-point constant (2.70)** of the four listed.
- The line for **pure water** on a vapour-pressure plot is the **highest** one (lowest curve number).
- Across the short periods, melting and boiling points **first increase then decrease** left to right.
- **Polyamide (nylon)** forms by condensing hexane-dioic acid with **hexane-1,6-diamine**.
- Most fluid flow is **turbulent rather than laminar because of high velocities**.
- Heat to change 1 kg solid → liquid at the melting point = **latent heat of fusion**.
- Water circulates in living organisms because of **cohesion and adhesion**.
- Newton's original formula **underestimated the speed of sound in air because he assumed isothermal** conditions.
- A sphere of radius r in a fluid of viscosity η: retarding force / speed = **$6\pi\eta r$** (Stokes).
- An oil film's colour pattern on water = **interference**.

Bled in from other units (same buckets)

- **Phenol reacts with NaOH but alcohols don't** — phenol is weakly acidic via **resonance stabilisation of the phenoxide ion**. →u16
- **Fischer esterification** is a **condensation** reaction. →u18
- A defective gene is transmitted to male offspring by the **female gamete**; viruses that infect bacteria are **bacteriophages**. →biology
- Real gases deviate at **low temperature and high pressure**. →u3
- Boiling water at 110 °C needs **760–1200 torr**. →u3

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u5 — Solids (crystal lattices, lattice energy, amorphous vs crystalline)

Unit cells and lattice types

- The **unit cell** = the smallest repeating part of a crystal lattice that still has all the properties of the whole crystal. ×3
- **Cubic** unit cell: **a = b = c, α = β = γ = 90°**.
- In a unit cell the angle **β is between faces b and c**.
- **NaCl** = **octahedral** arrangement of its crystal lattice. ×6
- **Iodine's** unit cell is **orthorhombic**; I_2 forms **molecular crystals**.
- The crystal's shape depends on the **arrangement of particles**.

Crystalline vs amorphous

- **Glass, rubber, plastics and charcoal are amorphous** — not crystalline. ×3
- **Copper, alum, sugar, NaCl, KCl, Fe, rhombic sulphur, copper sulphate** are crystalline.
- Amorphous solids **can possess small regions of orderly atomic arrangement**; they have no sharp melting point and do not cleave cleanly.
- Crystalline solids break along definite **cleavage planes**.
- **Copper crystals have no cleavage plane** (metallic, deforms rather than cleaves).
- **Graphite conducts better in one direction than another** because of **anisotropy**; properties that change with direction = **anisotropic** properties. ×2
- **Graphite is both anisotropic and a good conductor**.
- An element in different physical forms = **allotropy**; a compound in different crystal forms = **polymorphism**.

Lattice energy

- The **Born–Haber cycle** is used to evaluate **lattice energy**.
- **$Mg^{2+}(g) + O^{2-}(g) \rightarrow MgO(s)$** represents **lattice energy**.
- Lattice energy is the enthalpy of **formation** of the crystal from its gaseous ions.
- Lattice energy tells you about the **structure of ionic compounds**. ×2
- Lattice energy rises as ions get **smaller** and charges get **higher**:
  - **NaF > NaCl > NaBr > NaI** (F^- smallest → **NaF greatest**). ×2
  - **LiF** beats NaCl / KCl / CsI; **LiCl** beats NaCl / KCl / KBr.
  - **$C_6H_{12}O_6$** (a molecular solid) has the **lowest** lattice energy of the ionic list.
- Born–Haber steps: atomise the metal, ionise the metal, ionise the non-metal — "**delonize the metal**" is not a real step.

Conductivity in carbon allotropes

- **Graphite conducts** because **one valence electron per carbon is free to move** (delocalised between the layers); diamond does not.
- **Diamond is the covalent crystal** (NaCl, KI, $MgBr_2$ are ionic).

Bled in from other units (same buckets)

- In the ice lattice each O is attached to **four H atoms**. →u3
- **Polymorphism** in biology = *Obelia* (Coelenterata/Cnidaria) — the same word, different subject. →biology

- Amorphous-cell blood analysis is a damaged OCR line with no recoverable question.

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u6 — Chemical Equilibrium (Le Chatelier, Kc/Kp, buffers, solubility)

Equilibrium basics

- At equilibrium the **forward and backward rates are equal** — no net concentration change. ×2
- The ultimate fate of a reversible reaction = **a state with no net concentration change**.
- **Kc changes only with temperature** (not pressure or concentration). ×2
- A reaction proceeds **forward** when **Kc is large** (Kc = 10 beats 1, 1/10, 1/100); it goes nearly **to completion when Kc is large but positive**.
- Kc = 1 means neither side is favoured; Kc is never negative.
- Equilibrium expression **$[C]^2/[A][B]$** ↔ the equation **A + B ⇌ 2C**.
- **Kp = Kc when the moles of gas are equal on both sides** — e.g. **$H_2 + I_2 \rightleftharpoons 2HI$**. ×2
- **Kp > Kc when volume increases on the product side** (more gas moles made). ×6
- Direction of shift is predicted by **Le Chatelier's principle**. ×2
- Le Chatelier predicts the effect of **concentration, pressure and temperature** — it does **not** predict a change in **reaction rate**.
- Removing the **product** shifts equilibrium to the **product side**; raising pressure shifts to the **side with lower volume** (fewer gas moles). ×2
- Raising temperature in an **exothermic** reaction shifts equilibrium to the **reactant side**.
- **Kc increases with temperature only for endothermic reactions** — of the four listed, the papers key **$H_2 + I_2 \rightleftharpoons 2HI$**.
- Ozone decomposition at 25 °C has a huge **$K \approx 10^{55}$**.

Haber and Contact processes

- **Haber process** maximum ammonia yield = **high pressure, low temperature, continual removal of ammonia**. ×2
- The **NOT-true** way to raise NH_3 yield is **continuous addition of NH_3** (low temperature, high pressure and removal are all correct).
- **Decreasing temperature** raises ammonia yield (the reaction is exothermic).
- **200 atm** high pressure in Haber's process is for **better yield**.
- Haber's **catalyst = iron (Fe)**; nitrogen comes from **air**. ×2
- **Contact process** catalyst for sulfuric acid = **V_2O_5**.

Acids, bases and buffers

- Diluting a **weak acid with water increases $[H^+]$** (ionisation rises). ×6
- A **buffer resists change in pH** — e.g. **CH_3COOH + CH_3COONa**. ×2
- Heat of neutralisation of a strong acid + strong base ≈ **−57.4 kJ/mol**.
- A reaction in a **single step** = an **elementary reaction**.
- Rate rises with temperature because the **number of effective collisions increases**.

Solubility and Ksp

- Adding a common ion **decreases solubility**: $KClO_3$ solubility falls when **KCl** is added.
- **Ksp of $PbSO_4$ at 25 °C = 1.6 × 10^{-8} mol^2 dm^{-6}**.
- Salt AB (Ksp = 9 × 10^{-6}): molar solubility = √Ksp = **3 × 10^{-3}**.
- PbS (Ksp = 4.0 × 10^{-28}): ionic concentration = √Ksp = **2 × 10^{-14}**.
- $N_2 + O_2 \rightleftharpoons 2NO$, Kc = 0.1 at 2000 °C, 0.1 M each: **[NO] at equilibrium = 0.028 M**.
- $Ca(OH)_2$ (Ksp = 6.5 × 10^{-6}): solubility ≈ **1.17 × 10^{-2}**.
- **Ksp** = the product of ion concentrations in a saturated solution, each raised to its coefficient; for AgCl, **Ksp = $[Ag^+][Cl^-]$**.
- **Precipitation occurs when the ionic product is MORE than Ksp**; if it is **less than Ksp, no precipitation** occurs. ×2
- **Common ion effect** = the drop in solubility when a common ion is added — purifying NaCl by passing HCl gas, and precipitating AgCl by adding NaCl, are both common-ion effects. ×2

pH, pKa and buffers

- **pH + pOH = 14** for pure water at 25 °C.
- **Henderson–Hasselbalch: pH = pKa + log([base]/[acid])**.
- The buffer with the **highest pH** = the one with the **most conjugate base** (0.1 M CH_3COOH + **0.15 M** CH_3COO^-).
- **Higher Ka = stronger acid; lower pKa = stronger acid** (and higher pKa = weaker acid).
- The **weaker** acid of $HClO_4$ (−10) / HBr (−9) / H_2SO_4 (−3) / HCl (−7) = **H_2SO_4 (−3)** (highest pKa).
- **Kc has no units** when gas moles balance: **$H_2 + I_2 \rightleftharpoons 2HI$** and **$CO + H_2O \rightleftharpoons CO_2 + H_2$**.
- **$2HF \rightleftharpoons H_2 + F_2$**, Kc = 10^{-13}: **Kp = 10^{-13}** (equal gas moles, so Kp = Kc).

Haber and Contact detail

- Haber's process is **exothermic** and its equation is **$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$**.
- Haber optimum temperature = **400–450 °C** (modern plants 670–770 K, i.e. 400–500 °C).
- Haber catalyst = **iron crystals with metal-oxide promoters** (not MgO, Al_2O_3 or SiO_2).
- To raise NH_3 yield: **increase the pressure** (and remove ammonia, add excess N_2).
- **Contact process** $SO_2 \rightarrow SO_3$ catalyst = **vanadium pentoxide (V_2O_5)**.
- SO_3 is **not** absorbed directly in water because the reaction is **highly exothermic** (a mist forms); instead it goes to oleum.
- To raise SO_3 yield in **$2SO_2 + O_2 \rightleftharpoons 2SO_3$ (ΔH = −96)**: use **excess air** to drive the equilibrium right (and remove SO_3).
- **$2NO + O_2 \rightleftharpoons 2NO_2$ (ΔH = −114)** is favoured at **low temperature**.
- Le Chatelier gives **no information about time**.
- A **catalyst does not affect the equilibrium position** (only concentration, temperature and pressure do).

Bled in from other units (same buckets)

- **Esterification**: warming ethanol with ethanoic acid breaks the **O–H bond in ethanol** (the alcohol loses its hydroxyl hydrogen). →u18
- **Spring time period** (4 kg, k = 400 N/m): T = 2π√(m/k) = **0.628 s** — a physics item.
- **Highest lattice energy = LiF** — a u5 solids fact. →u5

u7 — Reaction Kinetics (rate laws, order, catalysts, collision theory)

Rate and its units

- **Rate of reaction = change in concentration ÷ time taken**. ×6
- Unit of rate = **mol dm^{-3} s^{-1}** (i.e. mol·dm^{-3}·s^{-1}). ×6
- Rate **decreases** as the reaction proceeds (reactants are used up). ×2
- The rate between two time intervals = the **average rate**; the rate at one moment = **instantaneous rate**.
- A **greater slope near the start** means a **greater** initial rate.
- Rate = k[A]^2[B]: halving [A] drops the rate by a factor of **4**.

Order and molecularity

- First order in A, second order in B → **Rate = k[A][B]^2**. ×5
- **S_n1 (unimolecular nucleophilic substitution) = first-order kinetics**. ×6
- **S_n2 = second-order kinetics**.
- When molar concentration of every reactant is **unity**, rate = the **rate constant** (they are equal). ×2
- The rate equals the **specific rate constant when concentrations are unity**.
- **Order of reaction from k's unit**: k in **s^{-1}** → **first order**; k in **L mol^{-1} s^{-1}** → **second order**. ×2
- Unit of the rate constant depends on the **order of reaction**. ×2
- **Rate constant depends on temperature** (not concentration, volume or pressure). ×2
- **Photochemical reactions** whose rate depends only on light intensity are **zero order**.
- **NO + O_3 → NO_2 + O_2** is **second order** (first in each reactant).
- The **rate-determining step** of a multi-step reaction is always the **slowest step**.
- The **Lucas test** is based on the **rate of reaction** (how fast the alkyl chloride appears).

Activation energy and collision theory

- **Activation energy** = the minimum energy molecules need to react on collision.
- For a reaction to occur, colliding molecules need **sufficient kinetic energy AND proper orientation** (collision theory).
- The unstable high-energy arrangement at the top of the barrier = the **activated complex / transition state**.
- Activation energy is **different for the forward and backward reaction**, and is **unaffected by a catalyst**.
- **Endothermic reactions have ΔH positive**.

Catalysts and enzymes

- A catalyst speeds a reaction by **lowering the activation energy** / providing an **alternative pathway** — it does **not** change ΔH, concentration of products, or shift equilibrium. ×2
- The **rate constant is independent of the initial concentration** of reactants.
- A substance that **retards** a reaction = an **inhibitor**.
- **Hydrogenation of oils** to solid fat uses a **nickel** catalyst.
- **Enzymes** are the biological catalysts; they are **organic, protein** catalysts that act **independently of the cell** — the false claim is that one enzyme catalyses **all** metabolic reactions.
- Enzymes lower activation energy by **providing an alternative pathway**.

- **Ligands act as Lewis bases** (they donate an electron pair).
- Raising **pressure** increases the rate only when **one or more reactants are gases**.
- The **Arrhenius equation** describes the effect of **temperature** on the rate constant.

Order — more definitions

- **Order of reaction** = the power to which a reactant's concentration is raised in the rate equation.
- The final equation expressing rate in terms of concentration = the **rate law**.
- **Zero-order** reactions: rate is **independent of the reactant concentration**. ×2
- A reaction whose rate doesn't depend on the concentration terms = a **zero-order reaction**.
- **Specific rate constant k is independent of concentration**; it is "specific" because it is the **rate per unit concentration**.
- Doubling [A] and quadrupling the product → **second order**.
- **2A + B → product** with B in large excess behaves as a **pseudo first-order** reaction.
- Photosynthesis **$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$** is a **zero-order** reaction (light-limited).
- Rate = k[A][B] with the slow step **$2NO + H_2 \rightarrow N_2 + H_2O_2$** matches the rate law **$k[NO]^2[H_2]$**.

Rate calculations

- First order, k = 0.0458 s^{-1}, [A] = 0.35 M → rate = **0.016 mol dm^{-3} s^{-1}**.
- First-order unit of k = **s^{-1}**.
- ΔC = 6 × 10^{-4} mol dm^{-3} over 10 s → rate = **6 × 10^{-5} mol dm^{-3} s^{-1}**.
- k = 0.693 min^{-1}, first order → half-life = 0.693/k = **1 min**.
- Raising temperature increases rate mainly because **collision frequency increases**.
- Forward rate falls with time because **reactant concentration decreases**.
- **Zn powder** gives the most H_2 with 1 M HCl (largest surface area).
- Rate of ionic reactions is studied by the **electrical conductivity method**.
- The **SN2 transition state** carries an overall **negative** charge.
- **Autocatalysis** = a product of the reaction acts as its own catalyst.
- **Activated-complex formation is endothermic** (energy is absorbed to reach the peak).
- **Enzymes increase rate by decreasing activation energy**; more substrate raises the rate **until all active sites are saturated**.
- The **Arrhenius plot** of log k vs 1/T is a **straight line with negative slope**.
- Pepsin's rate-vs-pH curve peaks near **pH 2** (a low-pH bell).

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u8 — Thermochemistry (first law, enthalpy, Hess's law, entropy, free energy)

First law and internal energy

- **First law = conservation of energy**; energy can be **converted from one form to another**, never created or destroyed. ×2
- Mathematical statement: **dE = q + w**. ×4
- Work done **by** the system and heat **absorbed** by it → **ΔE = q + (−w)** (the papers key B; note option A `q − w` is algebraically the same expression). ×6

- At **constant volume**, heat supplied = the **internal energy change**, **ΔE = qv**. ×2
- In an **isothermal** process the change in internal energy is **zero**.
- Raising internal energy can change state, allow a reaction, or raise temperature — **all of these**.

State vs path functions

- **State functions** depend only on the **initial and final states**: enthalpy, internal energy, pressure, volume, temperature, Gibbs energy.
- **Work is NOT a state function** (it is a path function). ×2
- **Intensive property = temperature** (mass, volume and enthalpy are extensive).
- The property that depends on the **amount of matter** = **Gibbs free energy** (extensive); density, pressure and temperature do not.

Enthalpy

- **ΔH = ΔE + PΔV** — enthalpy relates to **internal energy**. ×2
- **Exothermic**: heat flows **system → surroundings**, ΔH is **negative**. ×2
- **Endothermic**: ΔH is **positive**, and **ΔH(reactants) < ΔH(products)**. ×2
- **Boiling water is endothermic**.
- **ΔH°f of an element in its standard state = zero**; thermodynamically the most stable form of carbon is **graphite**.
- In **$2CO + O_2 \rightarrow 2CO_2 + X$ kJ**, X kJ is the **heat of reaction** (also the heat of combustion of CO).
- **Activation energy of an endothermic reaction is more than ΔH**.
- The **activated complex** has energy **higher than both reactants and products**.
- A **bomb calorimeter** measures the enthalpy of **combustion**.
- **Sabatier–Senderens** (hydrogenation) is **exothermic**.

Hess's law

- **Hess's law**: the enthalpy change of a reaction is the **same whether it happens in one step or several**. ×6
- Hess's law is an application of the **first law of thermodynamics** / the principle of **conservation of energy**. ×2

Entropy and free energy

- **Entropy (S)** measures the **randomness / disorder** of a system.
- At **chemical equilibrium ΔG = 0**.
- A **spontaneous process** occurs without continuous input of energy from outside.

Bled in from other units (same buckets)

- **Coordinate covalent bond** = one atom donates **both** electrons for sharing. →u10
- Cooling the reversible exothermic **NH_3** formation makes **more product (NH_3)** form. →u6

Spontaneity and reaction types

- **Spontaneous**: the **emission of α-particles from polonium** (N_2 + O_2, thermal decomposition of marble and burning paper all need a trigger).
- **Exothermic** among the set = **burning methane** (P_2O_5 decomposition, atmospheric NO formation and dry-ice sublimation are not).

- Most ordinary-temperature reactions with **−ΔH are exothermic**.
- **Reactants have higher energy than products in exothermic reactions**; **ΔH is negative** there.
- Heat flows **spontaneously from hot to cold**; a hot object **loses internal energy** when it warms a cold one.

Enthalpy definitions

- **Enthalpy of neutralisation** = enthalpy change when **1 mole of water** forms from acid + alkali under standard conditions.
- **Standard enthalpy of formation of CO_2 = −394 kJ/mol**.
- **Enthalpy of atomisation of hydrogen**: $\frac{1}{2}H_2(g) \rightarrow H(g)$, **ΔH = +218 kJ/mol**.
- In standard enthalpy of **atomisation** the surroundings' heat **increases** (energy is taken in).
- **Internal energy** = the sum of all forms of molecular energy in the system.
- **Volume is a state function**; ΔH is measured in **joules**.
- In an **isolated** system **neither heat nor mass** is transferred to the environment.
- Heat into a system with **no work done** → internal energy **increases**.
- 0.1 J into a gas expanding 0.1 m^3 at 1 N/m^2 (does 0.1 J of work) → internal energy **remains the same**.
- For a **cyclic** process the change in internal energy is **zero in both** systems.
- Internal energy of an ideal gas depends only on the **change in temperature**.
- **Adiabatic** first law: **Q = 0, so W = −ΔU**.
- With heat added and work done **by** the system: **ΔU = Q − W**.
- Thermodynamics studies the **relations between heat and mechanical energy**.

Hess's-law calculations

- ΔHf(CO) = −26.4, ΔHf(CO_2) = −94.0 kcal → heat of combustion of CO = **−67.6 kcal**. ×2
- $C + O_2 \rightarrow CO_2$ (−393.7); $CO + \frac{1}{2}O_2 \rightarrow CO_2$ (−283) → ΔHf of CO = **−110 kJ/mol**.
- Graphite ΔHc = −393.51, diamond ΔHc = −395.41 → graphite → diamond ΔH = **+1.91 kJ/mol**.
- $2H_2 + O_2 \rightarrow 2H_2O$ printed with ΔH = +285.5 is a sign typo in the source; the keyed option is **−205.5 kJ/mol**.

Bled in from other units (same buckets)

- **$2SO_2 + O_2 \rightleftharpoons 2SO_3$** heated shifts left: **SO_2 and O_2 increase, SO_3 decreases**. →u6
- Exothermic reactions are favoured by **decreasing temperature** (Le Chatelier). →u6
- STP = **14.7 psi and 0 °C**. →u3
- In a narrowing pipe an incompressible fluid speeds up to keep the **mass flow rate constant**, giving **high velocity, low pressure** (Bernoulli). →physics

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u9 — Electrochemistry (redox, cells, electrode potentials)

Redox fundamentals

- **Reduction** happens at the **cathode**; **oxidation** at the **anode** — in **both** galvanic and electrolytic cells. ×2
- In an **electrolytic** cell the **cathode is negative and reduction occurs** there.
- During **electrolysis, reduction always occurs at the cathode**.
- During **rusting**, iron behaves as the **anode** (it is oxidised).

- **$NO_2^- \rightarrow NO_3^- + ne^-$: n = 2** (N goes +3 → +5). ×2
- **Oxidation number of oxygen in OF_2 = +2** (fluorine is more electronegative). ×2
- **Chromium in $K_2Cr_2O_7$ = +6**.
- **Disproportionation** = the same element both increases and decreases its oxidation state.
- **Vanadium mostly acts as an oxidizing agent**. ×6
- **Photovoltaic cell** is the odd one out — the electrochemical cells are electrolytic, voltaic and (the solar cell being the same device as photovoltaic, so the papers key **D voltaic** as the intended answer; treat this item as flawed and learn the three real cell types). ×6

Oxidation states of the transition metals

- The first transition series' oxidation state rises up to **Mn** (+7 max), then falls.
- **Maximum oxidation state of Mn = +7**.
- **Iron's two common states = +2 and +3**.
- **Mn^{2+} has the $3d^5$ configuration** in its +2 state.
- **Zero oxidation state** appears in **$[Ni(CO)_4]$** (the metal carbonyl).
- **Phosphorus** commonly shows **+3 and +5**.
- **Toughness of transition metals** comes from their **binding energies**.

Cells and the salt bridge

- The **salt bridge transfers ions** and **maintains electrical neutrality** in the half-cells, completing the circuit and preventing charge imbalance. ×3
- **Fuel cells** convert a continuous fuel supply's chemical energy directly into **electrical energy**; the H_2–O_2 fuel cell's final product is **H_2O**.
- A **secondary (rechargeable) cell** = the **lead-acid accumulator** (dry cell and silver-oxide cell are primary).
- **Electrolysis of water**: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$ at the anode (and $2H^+ + 2e^- \rightarrow H_2$ at the cathode).

Electrode potentials and the series

- The **standard hydrogen electrode (SHE)** is assigned **0.00 V**.
- **E°red** of an element is measured by comparing it with the **SHE**.
- **Stronger the reducing agent, the more negative (smaller) its reduction potential**.
- Of P(−3.05), Q(−1.66), R(+0.34), S(+0.80) V, the **strongest reducing agent is P** (most negative).
- Metals **above hydrogen** in the electrochemical series have **negative** electrode potentials and displace H_2 from dilute acids.
- An element with a **more negative** reduction potential than hydrogen sits **above hydrogen** in the series.
- **Alkali metals are strong reducing agents** because of their **low ionization energies**.
- **Fluorine's high electronegativity** comes from its **small size and high nuclear charge** in its period.

Oxidation numbers — the ones that repeat

- **Free magnesium's oxidation number = 0** (an element in its free state is always 0).
- **Sulfur in H_2SO_4 = +6**; **$Na_2W_4O_{13}{\cdot}H_2O$: tungsten = +6**.
- **CrO_4^{2-} (+6), MnO_4^{2-} (+6), H_2SO_4 (+6), Na_2CO_3: carbon = +4** — the +4 species is **Na_2CO_3**.
- **$CH_3OH \rightarrow HCOOH$**: carbon goes **−2 → 0**.
- **$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$**: copper goes **0 → +2**.

- Balanced **Cu + H_2SO_4**: **$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$**.
- In all oxidation reactions atoms **lose electrons and increase their oxidation state**.
- **H_2 is the reducing agent** ($KMnO_4$, $K_2Cr_2O_7$ and O_3 are oxidising).
- **$K_2Cr_2O_7$** is the oxidising agent that turns acetaldehyde into acetic acid.
- The strong oxidising halogen oxyacid = **$HClO_4$** (perchloric).
- **Group IB (coinage metals)**: they have **positive reduction potentials**, lie **below SHE**, and **cannot displace H_2 from acids** — the false claim is that they are powerful reducing agents.
- SHE's **platinum provides a surface for oxidation and reduction** (it does not react or supply voltage).
- Connecting SHE to a Cu electrode: **Cu^{2+} is reduced** (copper sits below hydrogen).
- A clear **redox** example = **$H_2 + Cl_2 \rightarrow 2HCl$**; boiling water, dissolving NaCl and evaporating ethanol are not.
- **Not a redox reaction**: $CaCO_3 \rightarrow Ca + CO_3$ (no oxidation-state change).
- **Blast furnace $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$**: to make 56 g Fe you use **160 g** of hematite ore.

Electrolysis detail

- An **electrolytic cell uses direct current**, not alternating — "alternating current source is connected" is the wrong feature.
- In electrolysis, current is carried by **both positive and negative ions**.
- Electrolysing aqueous **$CuCl_2$** deposits **copper** at the cathode.
- **Galvanizing** an iron sheet coats it with **zinc**.
- A salt bridge **allows movement of ions between the two half-cells**.
- Balancing redox by the oxidation-number method does **not** require splitting into two half- reactions (that is the ion-electron method).

Bled in from other units (same buckets)

- **Water's central oxygen has 4 electron pairs** in its valence shell (2 bonds + 2 lone pairs). →u10
- **Maximum electrons in a sub-shell = 2(2l+1)**; the formula 2l+1 counts orbitals. →u2
- **n = 4, l = 3 (4f) has 7 orientations**. →u2
- **Cl^- has 8 electrons in its outermost shell**. →u2
- **β-particle emission**: the electron comes from the **nucleus**, not an orbit. →u20
- **Kα photon energy: hf = E_L − E_K** (an L electron falling into the K hole). →u15

More oxidation numbers

- **Cl in $MgCl_2$ = −1**; **S in SO_4^{2-} = +6**; **Mn in $KMnO_4$ = +7**.
- **Mn oxidation states** in $KMnO_4$, K_2MnO_4, MnO_2, $MnSO_4$ = **+7, +6, +4, +2**.
- The pair with the **same Fe oxidation state** = **$FeSO_4$ and $FeCl_2$** (both +2).
- **$2Na + Cl_2 \rightarrow 2NaCl$** is the redox reaction ($AgNO_3$ + NaCl is precipitation; the two half- reactions are not full redox equations).
- **$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$** is the **feasible** reaction by electrode potential (iron is above copper; Zn and Cd cannot displace Mg).
- In the SHE–Cu galvanic cell, current flows from the **copper electrode to the hydrogen electrode**.
- In **$2MnO_4^- + 10Cl^- + 16H^+ \rightarrow 5Cl_2 + 2Mn^{2+} + 8H_2O$**, **manganese is reduced from +7 to +2**.
- **Photochemical smog** = the build-up of oxidising agents (H_2O_2, HNO_3, PAN, ozone) in air.
- **Fe^{3+} has five unpaired electrons** in its ground state ($3d^5$).

- A **redox reaction is identified by transfer of electrons**.

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u10 — Chemical Bonding (VSEPR, hybridisation, MOT, periodic trends)

Shape and VSEPR

- **CH_4** bond angle = **109.5°** (regular tetrahedral). ×2
- **3 bonding pairs + 1 lone pair → trigonal pyramidal** (NH_3). ×2
- A triatomic molecule is either **linear (180°)** or **bent**. ×2
- **SF_6 is octahedral**; **PCl_5 is sp^3d** (trigonal bipyramidal).
- **NH_3 has a net dipole but BF_3 has zero** because **NH_3 is pyramidal** (BF_3 is symmetrical trigonal planar). ×2
- **NF_3's bond angle (102°) is smaller than NH_3's** because of fluorine's **high electronegativity** pulling bond pairs away from N. ×2
- **CO_2 has zero dipole moment** (linear, dipoles cancel); **CO has a non-zero dipole**. ×2
- **Water is liquid at room temperature** (unlike NH_3 and H_2S) because of **hydrogen bonding**. ×6
- **DNA's two chains are held/twisted together by hydrogen bonds**. ×2

Hybridisation

- **Benzene's carbons are sp^2**; the σ:π bond ratio in benzene is **6:1**. ×3
- **Phenol's –OH-bearing carbon is sp^2**. ×2
- **CH_3 (methyl radical) uses sp^2** hybridisation (NH_3, PH_3, SbH_3 are not sp^2). ×2

Bond strength and bond energy

- **Bond dissociation energy is highest for C≡C**. ×2
- **Highest H–X bond energy = HF**. ×2
- **Homolysis of a covalent bond gives free radicals**. ×2
- **Strongest metallic bond = Al** (most delocalised electrons); **calcium** has a strong metallic bond among K / P / Ca / S. ×2
- **H–F (ΔEN = 2.0) is a polar covalent bond**. ×2
- **NaOH contains both ionic and covalent bonds**.
- **Irreversible inhibition** = the inhibitor forms a strong **covalent bond** at the enzyme's active site. →u6

Molecular Orbital Theory

- **O_2 has 2 unpaired electrons** (MOT) — which is why it is **paramagnetic**; MOT explains this, VBT and VSEPR cannot. ×2
- Sigma and pi bonding are explained by **MOT and VBT**, not by VSEPR or CFT.
- **Coordinate covalent bond** = one atom donates **both** electrons.

Periodic trends

- **Smallest atomic radius** of Mg / S / P / Na = **S** (radius shrinks left→right across a period). ×3
- **Higher ionization energy → less metallic character** (and less electropositivity, less reducing power). ×2
- **Highest electron affinity of the halogens = chlorine** (not fluorine — F is too small).
- **Fluorine's high electronegativity** = small size + high nuclear charge.

- **Lower alkali metals react vigorously with water** because of **high electropositivity**.
- **Transition-metal compounds are coloured** due to **d–d electronic transitions**.
- **Born–Haber cycle → lattice energy of ionic solids**. →u5
- **Percent ionic character** for ΔEN = 4.0 − 1.20 = 2.8 → **0.73** (73%).

More shapes, angles and hybridisation

- **H_2O's bond angle (104.5°) is smaller than NH_3's (107°)** because water has **two lone pairs** repelling the bond pairs.
- **H_2S is bent / V-shaped**; **H_2O is distorted tetrahedral** (electron geometry tetrahedral, molecular geometry bent).
- **BF_3's ideal bond angle = 120°** (trigonal planar, **sp^2**).
- **NH_3's nitrogen is sp^3**; **CH_4 is sp^3** (BF_3, C_2H_4, $BeCl_2$ are not).
- **A C≡C triple bond = 1 sigma + 2 pi bonds**.
- **Shortest C–C bond length = C_2H_2** (triple bond beats benzene, ethene, ethane).
- **Ionization energy decreases down a group** (atomic and ionic radius and metallic character all increase).
- **Hydration energy depends on the charge-to-size ratio of the ion**.
- **Na^+ is smaller than a Na atom** because of the higher **effective nuclear charge**.
- **Smallest ionic radius = Al^{3+}** (highest charge, same shell).
- **Smallest shielding effect = Mg** (of Mg, Ca, Rb, Sr).
- **$CaCl_2$** is calcium chloride (Ca^{2+} + $2Cl^-$).
- **A very large electronegativity difference → predominantly ionic bond**.
- **CH_4's dipole moment = 0.0 D** (non-polar).
- **MOT: atomic orbitals adding form a bonding molecular orbital** (subtracting gives antibonding).
- **VSEPR explains molecular shapes** (not bond energy or length).
- **Bond length is NOT affected by the ionization energies** of the bonded atoms (it is affected by multiple bonds, hybridisation and electronegativity difference).
- **Secondary structure of a protein is set by hydrogen bonds**. →biology
- **Solid H_2 molecules attract by van der Waals forces**.
- **Energy changes in reactions come from both bond breaking and bond formation**.
- **Fluorine's electron affinity is less than chlorine's/bromine's** because of its **compact, dense small electron cloud**.
- **Positive electron affinity** (energy absorbed) = adding an electron to **O^-** (like charges repel).
- **H–H bond energy = 436 kJ/mol**.
- **Activation energy** = the minimum kinetic energy colliding molecules need to react. ×2
- **Kinetic theory of gases** = molecules in a gas are in constant movement. →u3
- **Coordination number** = the number of coordinate covalent bonds attached directly to the central metal ion. →u12

Ionization energy across a period

- **Increasing ionization energy of the third period: Na < Al < Mg < Si** (Mg's filled $3s^2$ pushes it above Al).
- **Highest ionization energy of Na / Mg / Al / Si = Si**; **lowest of Mg / Ar / Si / P = Mg**.
- **Smallest ionization energy of Li / Al / Ca / Ba = Ba** (falls down a group).

- **Lithium and beryllium's strong polarizing power** comes from their **small size** (high charge density).
- **$Mg(OH)_2$ precipitates from seawater** because its **solubility product is lower than $Ca(OH)_2$'s**.

Bonding detail

- **ΔEN < 1.7 → the bond is covalent**.
- **H_2S is angular with a non-zero dipole**; **BF_3 has zero dipole moment** (NF_3, NH_3, H_2O do not).
- **Non-polar covalent molecule = tetrachloromethane (CCl_4)**.
- **Ethene's carbons are sp^2 and coplanar**; **graphite's carbons are sp^2**.
- **Ethene is more reactive than ethane because of its weak, exposed pi bond**.
- **Aldehyde + HCN: the carbonyl carbon changes $sp^2 \rightarrow sp^3$**.
- **Maleic anhydride has 9 sigma and 3 pi bonds** (the papers key **9σ, 3π**).
- **Methyl chloride: 3 sigma bonds from sp^3–s overlap** (the C–Cl is sp^3–p).
- **Highest bond energy of the overlaps = sp–s**.
- **Smallest bond angle = H_2S** (of C_2H_2, NH_3, H_2S, $BeCl_2$).
- **1 lone pair + 3 bond pairs → trigonal pyramidal**.
- **Linear with 2 bond pairs**: CS_2, HCN, CO_2 — the exception is **$SnCl_2$** (bent).
- **Hybrid orbital size order: $sp^3 > sp^2 > sp$**.
- **Bond order = half the difference between bonding and antibonding electrons**.
- **Sigma/pi bonding is explained by MOT and VBT but NOT by VSEPR or CFT**.
- **VSEPR's false feature: "only lone pairs determine geometry"** — bond pairs count too.
- **Benzene's NOT-true statement: "no elimination reaction"** — it is sp^2, has fractional bond order and 6 equivalent monosubstitution sites.
- **Maximum polarizability = C_6H_{14}** (largest electron cloud).
- **CsCl with radius ratio 0.93 → body-centred cubic**.
- **Covalent radius of carbon in CH_3Cl = 176.7 − 99.4 = 77.3 pm**.
- **Nitrogen hybridisation in N_2Cl-type (N≡N–Cl) = sp**.
- **Shortest bond length (C–O vs C–N) = formamide** (C=O/C=N partial double bonds).
- **Lattice energy depends on the charge-to-size ratio**. →u5
- **NaCl lattice energy from Born–Haber = −727 kJ/mol**. →u5

Shapes, overlap and periodic trends — closing set

- **NH_3 is pyramidal**; **AX_2 with a 180° angle is linear**; **primary amines are pyramidal** at N.
- **Sigma bonds form by linear (head-on) overlap**; the sigma bond has maximum electron density **along the line joining the two nuclei**.
- **Diamond's C–C–C angle = 109°**; **methanol's carbon is sp^3**; **benzene's C–C bond length = 1.39 Å**.
- **Lattice energy is also called crystal energy**. →u5
- **KBr lattice energy from Born–Haber = −669.5 kJ/mol**. →u5
- **Paramagnetism comes from unpaired electrons**, and it **cannot be explained by hybridisation / VBT / VSEPR** — only MOT works. ×2
- **VSEPR repulsion order: lone-pair–lone-pair > lone-pair–bond-pair > bond-pair–bond-pair**; lone-pair repulsion is the **highest**.
- **Intermolecular repulsive forces in gases = van der Waals forces**. →u4

- **Electronegativity** = an atom's ability to attract the bonding electrons in a covalent bond.
- **Low ionization energy → loses electrons readily**.
- **Across a period the number of shells stays the same**; atomic radius falls because of the **increase in protons** (effective nuclear charge). ×2
- **Size order Li > Be** is correct (radius falls left→right).
- **Melting point drops sharply from Si to P** because of the **change in bonding and structure** (giant covalent → simple molecular).
- **Group IIA melts higher than IA** because IIA atoms **provide two bonding electrons**.
- **Ionic radius of F^- = 136 pm**.
- **In SiO_2 the Si:O ratio = 1:2**.
- **Average C–Br bond energy = 276 kJ/mol** (the papers key **290**).
- **In aldehydes/ketones the sigma bond of C and O is the one distorted toward oxygen**.
- **Antibody heavy + light chains are bonded by disulphide bonds**. →biology
- **Water vapourises by breaking hydrogen bonds**; its high heat of vaporisation regulates heat. →u4
- **H atom n=4 → n=2 photon = 2.55 eV**. →u2
- **Maximum probability of finding the H electron = 0.053 nm** (the Bohr radius). →u2
- **Vinyl acetylene IUPAC = but-1-en-3-yne**. →u14
- **Benzene is NOT highly reactive** — it is aromatic, stable and obeys Hückel's rule. →u14
- **Starch and cellulose are both glucose polymers**. →biology
- **Lipase** breaks lipids into fatty acids. →biology
- **Dihaloalkane → alkyne involves elimination, base and heat — not addition**. →u15

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u11 — s- and p-Block Elements

Group I A — alkali metals

- **Metallic character increases down the group** (both alkali and alkaline-earth metals). ×6
- **Most reactive with water = cesium**; **slowest-reacting = lithium**. ×2
- **Lithium differs from the other alkali metals** because of its **small radius and high charge density**.
- **Lithium is the alkali metal that forms only the normal oxide (Li_2O)** with O_2 (Na gives the peroxide, K/Rb/Cs the superoxide).
- **Alkali-metal chlorides dissolve in water to a neutral pH 7 solution**.
- **Alkali metals have the highest atomic radius in their period**.
- **Flame colours: Li = red**, Na = golden yellow, K = lilac, Cs = blue-violet.
- **Alkali metals react slowly with oxygen** compared with the alkaline earths.

Group II A — alkaline earth metals

- **Group IIA = alkaline earth metals**.
- **Beryllium oxide (BeO) is amphoteric** (CaO, K_2O, Li_2O are basic).
- **Be reacts with oxygen to form a BeO layer that protects the metal from further oxidation**.
- **Strontium** (between Ca and Ba) is predicted to form a **sparingly soluble sulfate**.
- **Plaster of Paris = $CaSO_4 \cdot ½H_2O$** (gypsum is the dihydrate).

p-block and general

- **Majority of periodic-table elements are metals**.
- **Most electronegative element = fluorine**.
- **Group VIIA = halogens**; **VIA = chalcogens**; **VA = pnictogens**.
- **Borax is a compound of boron**.
- **Iodine is the halogen that is solid at room temperature and sublimes**.
- **Helium is the noble gas** of the four listed; **noble-gas inertness = completely filled valence shells**.
- **Argon provides the inert atmosphere for welding**.
- An **acid** gives H^+ in water.
- **Friedel–Crafts alkylation**: benzene + isopropyl chloride / $AlCl_3$ → isopropylbenzene. →u14
- **Hydration of propene with H_2SO_4 gives propan-2-ol** (Markovnikov). →u14
- **Chlorobenzene + NaOH at 350 °C, 150 atm → phenol** (Dow process). →u16
- A **base-catalysed reaction is nucleophilic**. →u13
- **NaCl, aluminium oxide and iron oxide** have characteristic heats of fusion — **silicates** are the exception.
- **Casein glue is the natural adhesive** (silicones, PVA and polyamide are synthetic). →u15

Halogens

- **Oxidising power F > Cl > Br > I**, so the possible reaction is $Br_2 + 2I^- \rightarrow I_2 + 2Br^-$.
- **Lowest X–X dissociation energy = F_2** (lone-pair repulsion in the tiny molecule).
- **Alkenes react with hydrogen halides in the order HI > HBr > HCl**. →u14

Nitrogen, sulfur and their oxoacids

- **N_2 is very less reactive** because of its **strong triple bond**.
- **Ammonia solution is basic**.
- **First ionisation of H_2SO_4: $H_2SO_4 + H_2O \rightarrow H_3O^+ + HSO_4^-$**.
- **SO_3 absorbed in concentrated H_2SO_4 gives oleum**.
- **The anhydride of sulphuric acid = SO_3 (sulphur(VI) oxide)**.
- **Unpolluted rain is slightly acidic** from **carbon dioxide** (carbonic acid).
- The **fawn-brown colour of photochemical smog = nitrogen dioxide**. →u9
- **Oil of vitriol = sulfuric acid**.

Ozone layer

- **Ozone concentration is measured in Dobson units**.
- **Ozone sits 10–50 km above Earth** and absorbs UV.
- **CFCs deplete/thin the ozone layer**, so ozone destruction raises **ultraviolet radiation**.

Alkaline earth reactions

- **Alkaline earth metals + nitrogen → M_3N_2**.
- **Alkaline earth hydroxides decompose: $M(OH)_2 \rightarrow MO + H_2O$**.
- **Least soluble alkaline-earth oxide = MgO** (solubility rises down the group).
- The **oxide layer on an alkaline-earth metal protects it from further atmospheric attack**. →u11

Bled in from other units (same buckets)

- **0.5 mol NaOH = 20 g**. →u1
- **Mostly p-block elements form binary compounds with oxygen**.
- **Ethyl bromide + excess ammonia → ethylamine ($CH_3CH_2NH_2$)**. →u18
- **Ethanol + $Na_2Cr_2O_7$ / dilute H_2SO_4, distilled immediately → acetaldehyde** (full oxidation would give acetic acid). →u17
- **The acid that reacts with $NaHCO_3$, NaOH and Na_2CO_3 = propanoic acid (CH_3CH_2COOH)**. →u18
- **Saliva contains water, mucus, amylase and sodium bicarbonate**. →biology

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u12 — Transition Elements and Coordination Chemistry

The 3d series

- **Atomic numbers of the 3d transition series = 21–30**. ×6
- **Iron is the typical transition element** (Ca, Al and Pb are not).
- **Unpaired electrons increase up to groups VB and VIB** (Cr/Mn peak), then fall.
- **Binding energy weakens progressively up to group IIB**.
- **Transition-metal compounds with unpaired electrons are attracted by a magnet** (paramagnetic).
- **d-block elements are entirely metallic** (s-block is too, so "both A and C").

Alloys and interstitial compounds

- **Transition elements form substitutional alloys easily because their atoms are nearly the same size**. ×2
- **Group IB elements usually form alloys**.
- **Brass = copper + zinc**; **bronze** and **steel** are alloys too; **graphite is NOT an alloy**.
- **Interstitial compounds** form because of the **holes in the metal crystal lattice**.
- **Chelates are more stable than ordinary (monodentate) complexes**.

Coordination chemistry

- The **central metal ion + its ligands = the coordination sphere**.
- **Coordination number** = the number of coordinate bonds to the central metal. →u10
- **Polydentate / chelating ligands** bind through more than one donor atom; **chelate** comes from the word for **crab's claws**.
- **Diethylenetriamine is tridentate**.
- **Bidentate ligands: hydrazine (N_2H_4) and oxalate ($C_2O_4^{2-}$)**; monodentate: ammine, aqua, carbonyl, Br^-, CN^-, OH^-.
- **NH_2^- (amide/amine ligand) is an anionic ligand**.

Chromium chemistry

- **Potassium chromate + acid → potassium dichromate**.
- **$K_2Cr_2O_7$ / H_2SO_4 oxidises ethanol to acetic acid (CH_3COOH)**. →u16
- **Propanal** does not react with sodium metal but oxidises (reflux, acidified dichromate) to a carboxylic acid. →u17
- **Aldehydes are NOT easily oxidised by HCl** (they are by $K_2Cr_2O_7$ and $KMnO_4$). →u17

- **1-pentyne vs 2-pentyne: $AgNO_3$ + NH_4OH** (terminal alkyne precipitates). →u14
- **Fischer esterification (CH_3COOH + C_2H_5OH) is catalysed by H_2SO_4.** →u18

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u13 — Fundamental Principles of Organic Chemistry (nomenclature, isomerism)

IUPAC names that repeat

- **CH_3–$CH(Cl)$–CH_2–CH_2–CHO = 4-chloropentanal** (number from the –CHO carbon). ×6
- **Malonic acid $CH_2(COOH)_2$ = 1,3-propanedioic acid**. ×6
- **CH_3–CH=CH–CH_2–C≡CH = 4-hexen-1-yne** (the triple bond gets the lower number). ×6
- **Neopentane = 2,2-dimethylpropane**. ×6
- **$(CH_3)_3CCH_2Br$ = 1-bromo-2,2-dimethylpropane**. ×2
- **CH_3–CH_2–CO–CH_2–CHO = 3-oxopentanal**.
- **Stearic acid = octadecanoic acid**; **chloroform = trichloromethane**.
- **α,β-dimethylvaleric acid = 2,3-dimethylpentanoic acid**.
- **Diisopropyl ketone = 2,4-dimethylpentan-3-one**.
- **2-Bromo-4-chloro-3-iodopentane** is the correct alphabetical/lowest-locant name.

Isomerism

- **Fumaric and maleic acid are geometrical (cis-trans) isomers**. ×6
- **Maleic acid** is the one that shows cis-trans isomerism (of malonic, succinic, lactic).
- **1-chloropropane and 2-chloropropane = position isomers**. ×3
- **Enantiomers are optical isomers**.
- **Vinyl alcohol and acetaldehyde are tautomers**; **tautomerism = shifting of a proton within the same molecule**. ×2
- **Metamers differ in the distribution of alkyl groups around the functional group**.
- **$CH_3CH_2CH_2CH_2COCH_3$ (a ketone) is the functional isomer of the aldehyde $CH_3CH_2CH_2CH_2CH_2CHO$**.
- **A molecule is chiral if it has no element of symmetry** (non-superimposable mirror image). ×2
- **$[Co(en)_3]^{3+}$ exhibits optical isomerism**.

Classification and functional groups

- **Saturated alicyclic hydrocarbons = CnH_2n**. ×6
- Organic compounds split into **open chain and closed chain**; **homocyclic** rings divide into **alicyclic and aromatic**. ×2
- **Acid halides = R–COX**; **esters = R–CO–OR**; **ethyl acetate has the ester group**.
- **Amino acids carry amino + carboxyl groups**; natural ones are the **L-form**; **glycine is the simplest and has no chiral centre**. ×2
- **Chlorophyll a's functional group = $–CH_3$**.
- **Catalytic cracking** gives the better-quality gasoline. →u20

More nomenclature

- **$CH_3CH_2COCH_2CHO$ = 3-oxopentanal**.

- **$C_6H_5OCH_3$ = methoxybenzene**.
- **$CH_3CH_2CH_2C{\equiv}CH$ = pent-1-yne**; **CHCCl = chloroethyne**.
- **$Br–CH_2–CH_2–CO–CH_3$ = 1-bromobutan-2-one**.
- **$CH_3(CH_2)_4CH(CH_3)_2$ = 2-methylheptane**.
- **$C_2H_5CH=C(C_3H_7)C_2H_5$ = 4-ethylhept-3-ene**.
- **Vinyl acetylene = but-1-en-3-yne**.
- **$CH_3CH_2CH_2CHO$ = butanal**; **isopropyl alcohol = 2-propanol**.
- **Ketone's functional group = carbonyl**.

More isomerism

- **Diethyl ether and methyl n-propyl ether are metamers**.
- **2-pentanone and 3-pentanone are position isomers**.
- **(+) and (−) tartaric acid are enantiomers**.
- **2-hydroxypropanoic (lactic) acid shows optical isomerism** (it has a chiral carbon).
- **Pent-1-ene does NOT show geometrical isomerism** (a terminal double bond has two identical H's on C1); but-2-ene, 1,2-dimethylcyclopropane and 1,3-dimethylcyclopentane do.
- **3-methyl-2-pentene** is the alkene that shows geometrical isomerism.
- **1-butene and 2-butene are position isomers**.
- **Geometrical isomerism** = same structural formula, different position of atoms across a carbon–carbon bond.
- **Metamerism is NOT a type of stereoisomerism** (conformational, optical and geometrical are).
- **Pyridine is heterocyclic**.
- **Highest atomic radius of group II = barium**. →u11

Analysis and miscellany

- **Oxygen cannot be detected directly** in an organic compound by the usual qualitative tests (Cl, N and P can).
- **Biogas is not a fossil fuel** (coal, petroleum and natural gas are). →u20
- For **2A + B → 3C + D**, the rate is **NOT** expressed as **$-d[C]/dt$** with a wrong sign or a missing coefficient — the wrong expression is **$-1/3\ d[C]/dt$** (product rate is positive).
- **Transition-element ions and compounds are NOT colourless** (they are coloured by d–d transitions). →u12
- **Homocyclic compounds divide into alicyclic and aromatic**.

Final nomenclature and isomerism set

- **Formaldehyde = methanal**; the aldehyde-series member **H–CHO = methanal** (acetaldehyde is ethanal, propionaldehyde is propanal).
- **$CH_3–CH_2–C(Cl)(CH_3)–CH_2–CH_3$ = 3-chloro-3-methylpentane**.
- **$CH_3–CH(CH_3)–CH_2–CH(CH_3)–CH_3$ = 2,4-dimethylpentane**.
- **$CH_2=CH–CH=CH–CH=CH–CH_3$ = hepta-1,3,5-triene**.
- **$CH_3–CH(CH_3)–CH_2Cl$ (isobutyl chloride) = 1-chloro-2-methylpropane**.
- **Metamerism = unequal distribution of alkyl groups on the two sides of a carbon atom**.
- **Benzene, naphthalene and anthracene are aromatic hydrocarbons**.

- **cis-trans pairs**: the two **CH_3–CH=CH–CH_3** drawings are cis/trans isomers; **2-butene** shows it, and the cis isomer is the one with both groups on the same side.
- **3,4-dimethyl-3-hexene** is the skeletal-formula hydrocarbon's IUPAC name.
- **Early chemists believed organic compounds could only be made by and within living things** (vitalism).
- **A sharp melting point = a crystalline substance**: **diamond** (glass, coal tar and gemstone mixtures do not). →u5
- **Carbon tetrachloride is used as a solvent**.
- **The aspartic-acid/phenylalanine dipeptide (aspartame) shows 4 functional groups**.

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u14 — Chemistry of Hydrocarbons (alkanes, alkenes, alkynes, benzene)

Benzene and aromaticity

- **1,3,5-cyclohexatriene** is the reference used to calculate benzene's stability (resonance energy). ×6
- **Greater resonance energy → higher stability**. ×2
- **Electrophilic substitution on benzene needs a Lewis acid catalyst** ($AlCl_3$, $FeBr_3$). ×6
- **Benzene's C:H mass ratio = 12:1**. ×6
- **Benzene on oxidation gives maleic anhydride**.
- **The most volatile liquid of water / methanol / benzene / diethyl ether = diethyl ether**. ×2

Benzene substitutions

- **Toluene + Br_2 / UV → benzyl bromide** (side-chain free-radical substitution). ×2
- **Toluene + Cl_2 / Lewis catalyst → equal mixture of ortho- and para-chlorotoluene**. ×2
- **Ring deactivators: –NO_2 (strongest), –COOH, –Cl (deactivating but o/p-directing); –NH_2 is an activator**.

Alkanes

- **Free-radical substitution of ethane + halogen needs heat and UV light**. ×2
- **The three pentane isomers: n-pentane, 2-methylbutane (isopentane), 2,2-dimethylpropane (neopentane)**.
- **Neo-** is the prefix for two methyls on the second-last carbon.
- **Alkanes have only single (sigma) bonds**. →u2

Alkenes

- **The characteristic alkene reaction is electrophilic addition**. ×2
- **1-butene and 2-butene are position isomers**. →u13

Alkynes

- **Alkynes are prepared by dehydrohalogenation of a vicinal dihalide**.
- **Terminal alkyne + H_2 / Lindlar catalyst → cis-alkene** (the papers key cis; a genuinely terminal alkyne gives the terminal alkene). ×2

Bled in from other units (same buckets)

- **Potassium forms the superoxide (KO_2)** with oxygen; Li the normal oxide. →u11
- **Strongest base of the hydroxides = NaOH** (of Ca, Li, Mg, Na). →u11

- **LiOH is the heat-stable alkali-metal hydroxide** (the papers key Li). →u11
- **Atomic solids = covalent solids**. →u5
- **Metal conductivity falls with temperature because metal-ion oscillations hinder electron movement**. →physics
- **Most common coordination number = 6**; **most abundant transition metal in the crust = iron**; **coinage metals = Cu, Ag, Au**; **paramagnetism = unpaired electrons**; **haemoglobin's central metal = iron**. →u12
- **Strontium burns with a brick-red flame**. →u11
- **Excess ethanol + conc. H_2SO_4 at low temperature → diethyl ether**. →u16
- **Metallic property increases down a p-block group**. →u11
- **Mycorrhizae = mutualism**. →biology

Alkanes — reactions

- **Chlorination of methane with excess Cl_2 / UV → carbon tetrachloride** (the final product).
- **The chlorine-radical attack on methane is the propagation step** of free-radical substitution.
- **n-butane + $AlCl_3$/HCl → isobutane** (isomerisation).
- **Alkanes cannot undergo addition** (they do substitution, combustion and cracking).
- **Least reactive hydrocarbon = benzene**; **reactivity order: alkenes > alkynes > alkanes**.
- **Combustion of methane is NOT endothermic** (it is exothermic; water decomposition, ethane dehydrogenation and graphite→diamond are endothermic).
- **3.2 kg methane needs 12.8 kg O_2** for complete combustion ($CH_4 + 2O_2$).
- **1.6 g methane contains 6.022×10^{23} electrons** (0.1 mol × 10 e^-).

Alkenes — reactivity

- **Most reactive toward bromine addition = 2-methyl-2-butene** (most alkyl-substituted).
- **Reactivity toward hydrogen halides: 2-methylpropene > propene > ethene** (more substituted = more reactive).
- **2-bromopropane + sodium ethoxide → propene** (elimination).

Benzene — more

- **Benzene is made in the lab from sodium benzoate + NaOH/CaO** (soda lime decarboxylation).
- **Acetophenone from benzene by acylation** (Friedel–Crafts).
- **Alkylbenzene from benzene + alkyl halide / anhydrous $AlCl_3$ = Friedel–Crafts alkylation**.
- **Delocalised electrons in benzene move freely in the π system above and below the ring plane**.
- **Benzene + 3 Cl_2 → benzene hexachloride proves benzene has 3 double bonds**.
- **Electrophilic aromatic substitution = a proton (H^+) on the ring is replaced by an electrophile**.
- **$–NH_3^+$ (ammonium) deactivates the ring**; **$–NH_2$ activates it**.
- **To make 2-chloromethylbenzene: $CH_3Cl/AlCl_3$ first, then Cl_2 in diffused sunlight**.
- **Anthracene = three linearly fused benzene rings** (phenanthrene is angular).
- **1,3-benzenediol = resorcinol** (1,2 = catechol, 1,3,5 = pyrogallol).
- **Butanal forms a crystalline phenylhydrazone** (aldehydes/ketones do; esters and alcohols don't). →u17

Bled in from other units (same buckets)

- **Triclinic crystal: all axes and angles unequal**. →u5

- **Metallic thermal conductivity is due to freely moving electrons; Na's coordination number in metallic sodium = 12**. →u5, →u12
- **Sodium burns with a golden flame and reacts vigorously with cold water**. →u11
- **Aluminium is extracted from bauxite**. →u20

Benzene reactions — mechanism detail

- **Sulphonation's electrophile = SO_3**.
- **Chlorination of benzene with $FeCl_3$ = electrophilic substitution; nitration of benzene is the classic electrophilic substitution**.
- **Toluene → p-chlorotoluene uses Cl_2 + $AlCl_3$** (Lewis catalyst, ring substitution).
- **Most readily sulphonated = toluene** (the methyl group activates the ring).
- **Electron-releasing ring substituent = $–NH_2$** ($–NR_3^+$, –CN and $–NO_2$ withdraw).
- **Friedel–Crafts acylation catalyst = $AlCl_3$**.
- **Benzene is NOT highly reactive** (aromatic, stable, obeys Hückel's rule).
- **Benzene's density = 0.88 g/cm³**.

Alkynes

- **Acetylene + HBr → 1,1-dibromoethane** (Markovnikov geminal dihalide).
- **Terminal alkynes are acidic because the terminal carbon is sp hybridised** (50% s-character stabilises the acetylide).
- **Dihaloalkane → alkyne needs elimination, base and heat — not addition**.

General hydrocarbon facts

- **Reactivity order: alkenes > alkynes > alkanes**.
- **Ethene is the unsaturated hydrocarbon** (ethane, propane, butane are saturated).
- **Alkenes and alkynes have double/triple bonds between adjacent carbons**.
- **Hydrogenation of alkenes uses Ni, Pt or Pd — all of these**.
- **Homolytic C–H fission in an alkane gives an alkyl free radical**.
- **Markovnikov's rule**: in an unsymmetrical alkene the **negative** part adds to the double-bond carbon with the **fewer hydrogens**.
- **Natural gas is mainly methane**.
- **2-butene gives only one aldehyde on ozonolysis** (acetaldehyde). →u17
- **Ester from 2,2-dimethylpropanoic acid + ethanol**. →u18

Bled in from other units (same buckets)

- **Alkaline-earth sulfates decrease in solubility down the group**. →u11
- **$BeCO_3$ decomposes most readily on heating** (carbonate stability rises down the group). →u11
- **Iron in $CuSO_4$ solution: copper is deposited** (Fe is more reactive). →u9
- **18.5 g Al → 34.9 g Al_2O_3**. →u1
- **Alcohol + sodium → alkoxide** (+ H_2). →u16

Alkanes — bonding and naming

- **Ethane has 7 bonds** (1 C–C + 6 C–H).

- **Alkanes have only single bonds between carbons**.
- **Removing an H from an alkane gives an alkyl group** (aryl/phenyl come from benzene).
- **Alkene general formula = CnH2n**.

Benzene — named reactions

- **Benzene + Cl_2 in sunlight → hexachlorobenzene** (addition, not substitution).
- **Nitration's electrophile = NO_2^+**.
- **Sulphonation's electrophile = SO_3; chlorination's powerful electrophile = Cl^+**.
- **Substituting –H by $–NO_2$ = nitration; introducing –CO–R = acylation; introducing an alkyl group (with $AlCl_3$) uses R–Cl (chloroalkane)**.
- **Ethane + Br_2 intermediate = the ethyl free radical with Br ($CH_3CH_2\cdot$ + Br·)**.
- **Propene + HBr → 2-bromopropane ($CH_3CHBrCH_3$)** (Markovnikov).
- **Alcoholic KOH/NaOH + halogenoalkane → alkene by elimination (dehydrohalogenation)**.
- **Hydrogenation of unsaturated oils uses finely divided nickel**. →u8

Bled in from other units (same buckets)

- **Coinage metals are least reactive because of their positive reduction potential**. →u11
- **Metallic conduction = free movement of electrons through the lattice**. →u5
- **X-rays are produced by bombarding electrons on a target metal**. →u15
- **Parallel-wire currents in opposite directions repel**. →physics
- **Resistors**: 7 × 2 Ω giving 10/11 Ω = **5 parallel + 2 series**; a resistor R cut into 3 and paralleled = **R/27**... the keyed option is **R/9** (each third is R/3, three in parallel = R/9). →physics
- **Ethylene glycol is an anti-freezing agent**. →u16
- **CH_3COOH + Mg → magnesium acetate**. →u18
- **Aluminium clogs fish gills and suffocates them**. →biology
- **Aerobic decomposition of glucose gives methane**? No — aerobic gives CO_2 + H_2O; the keyed answer here is **methane** (anaerobic). →biology
- **Burning plastic waste is used as an industrial fuel / for electricity**. →biology
- One corpus line here is an English sentence fragment (population-growth paradox) with no recoverable question.

Biology that landed in this unit (recorded for coverage; revise from the Biology notes)

- **Biological control** of pests = regulation by natural enemies, predators, parasites and pathogens.
- **Cyanobacteria are photoautotrophic bacteria**.
- **Viruses are obligate intracellular parasites**; **Ascaris lumbricoides** is the common **intestinal roundworm of humans and pigs** (phylum Nematoda).
- **Radial symmetry** represents a **sessile** mode of life.
- The **menstrual cycle is regulated by gonadotrophins** (FSH/LH), not plant hormones.
- **Mutualism / symbiosis** = both partners benefit; **commensalism** = one benefits, the other is unaffected; **parasitism** = one benefits, one is harmed.
- **Pheretima posthuma** is the earthworm; **Hirudo medicinalis** is the leech.

Physics that landed in this unit

- **Photoelectron emission depends on the frequency of the incident light**.
- The rule for induced current direction is **Lenz's law**.
- Ease of reduction of H^+ / Cu^{2+} / Ag^+: **Ag^+ > Cu^{2+} > H^+** (silver is the easiest to reduce).
- **Boron has 1 unpaired electron** in its ground state ($2p^1$). →u2
- **Carboxylic acid + ammonia → ammonium salt** on heating gives an **amide**. →u18

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u15 — Alkyl Halides (SN1, SN2, E1, E2)

Substitution mechanisms

- **SN2**: C–X breaks and C–Nu forms **simultaneously** (one step, backside attack, complete inversion). ×2
- **SN1**: goes through a **carbocation intermediate**; gives **50% inversion + 50% retention** (racemisation); its rate depends on the **concentration of substrate only**. ×2
- **SN1 reactivity: tertiary > secondary > primary** (stable carbocation).
- **SN2 reactivity: primary > secondary > tertiary** (least steric hindrance).
- **E1 is two steps; E2 is one step**.
- **Strong bases favour elimination over substitution**; **lower temperature favours substitution**; the leaving group matters in both.

Leaving group and reactivity

- **Reactivity of alkyl halides: R–I > R–Br > R–Cl > R–F** (weakest C–X bond = most reactive). ×3
- **Least reactive = R–F**; **most reactive = the iodo-compound** (lowest bond energy).
- **Ascending order of reactivity: F < Cl < Br < I**.
- **Dehydrohalogenation uses alcoholic KOH** (aqueous KOH gives substitution/alcohol). ×2
- **2-bromobutane + alcoholic KOH → 2-butene as the major product** (Saytzeff).

Classifying alkyl halides

- **2-bromo-2-methylbutane = tertiary**; **isopropyl chloride = secondary**.
- **Carbocation** = a carbon with a positive charge attached to three groups.
- **Nucleophile** = a species with an unshared electron pair (**H_2O is one**; $AlCl_3$, BF_3 are electrophiles/Lewis acids).
- **Acylium ion is an electrophile**.

Named reactions

- **Wurtz reaction**: alkyl halide → alkane with sodium in dry ether.
- **Grignard reagent RMgX is prepared in ether** (never aqueous — it reacts with water and with ethanol).
- **Alkyl halides → ethers when heated with Ag_2O / moist silver oxide**.

Bled in from other units (same buckets)

- **Carbonyl reactivity toward nucleophiles: H_2CO > RCHO > R_2CO** (less steric hindrance). ×6 →u17
- **Electron-withdrawing substituents increase phenol's acidity**; **phenol is more acidic than ethanol because its conjugate base is resonance-stabilised**. ×2 →u16

- **Pentan-2-one + HCN gives a chiral product** (cyanohydrin at a new stereocentre). →u17

Nucleophiles and electrophiles

- **Nucleophiles: NH_3, H_2O, CN^-, NH_2^-, CH_3^-, Cl^-** (they carry a lone pair or negative charge).
- **NOT a nucleophile = BF_3** (it is a Lewis acid / electrophile, as are NO_2^+ and NH_4^+).
- In **HCHO + HCN → cyanohydrin**, the attacking nucleophile is **CN^-**.
- In **$CH_3COOH + PCl_5$**, the nucleophile that attacks the acid carbon is **Cl^-**.
- **Secondary / dihaloalkanes can go by both SN1 and SN2**.
- **CH_3CH_2Br + alcoholic KOH → ethene follows the E2 mechanism**.
- **Reactivity toward nucleophiles: RI > RBr > RCl > RF**.

Haloform applications

- **Chloroform is the suspected liver carcinogen with negative reproductive/developmental effects**.

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u16 — Alcohols and Phenols

Alcohols — reactions

- **Oxidation of a secondary alcohol → ketone; primary → aldehyde → carboxylic acid**. ×4
- **Dehydrogenation converts ethyl alcohol into acetaldehyde**.
- **Secondary propyl alcohol (propan-2-ol) oxidises to acetone**.
- **Dehydration of an alcohol to ethene is an elimination reaction**. ×2
- **Dehydration at 140 °C with H_2SO_4 gives diethyl ether** (intermolecular); at 170–180 °C it gives ethene.
- **Ketones + $LiAlH_4$ in dry ether → secondary alcohols; aldehydes → primary alcohols**.
- **Ketone + Grignard reagent → tertiary alcohol**.
- **Acetaldehyde + ethanol + acid first gives a hemiacetal** (then the acetal).
- **Ethanol gives the iodoform reaction** (it has the CH_3–CH(OH)– group); methanol, 1-butanol and 1-propanol do not.
- **Methanol and ethanol are both primary alcohols**.
- **Glycerol = 1,2,3-propanetriol; glycol = the 1,2-diol**.
- **Glycerin inhibits the decomposition of H_2O_2** (MnO_2 and V_2O_5 catalyse it).

Alcohols — acidity and the Lucas test

- **Acidity of alcohols: primary > secondary > tertiary**.
- **C–O bond cleavage reactivity: tertiary > secondary > primary**. ×2
- **Lucas reagent distinguishes primary, secondary and tertiary alcohols**; the **most reactive is 2-methyl-2-butanol** (tertiary). ×2
- **The "oily layer" on heating = 2-methyl-2-butanol** (the tert-alcohol forms the insoluble chloride).
- **The alcohol-vs-phenol differentiating test is the Lucas test**.

Phenols

- **Phenol + concentrated HNO_3 → picric acid** (2,4,6-trinitrophenol); **3 mol conc. HNO_3 with H_2SO_4 → picric acid**. ×2

- **Phenol is more reactive than benzene** toward electrophilic substitution (–OH activates).
- **Phenol is more acidic than water and ethanol but less acidic than a carboxylic acid** — its conjugate base is resonance-stabilised. ×2
- **The alkoxide from deprotonating an alcohol acts as a Lewis base**. ×6
- **Bakelite = phenol + HCHO polymerised in dilute base**.
- **Picric acid and ascorbic acid do NOT contain a –COOH group** (they are not carboxylic acids).

Bled in from other units (same buckets)

- **Methanal (formaldehyde) is the sp^2 carbon** among methanal / methanol / ethanol / propan-2-ol. →u17
- **2-bromobutane + alcoholic KOH → 2-butene**. →u15
- **Chlorobenzene + NaOH at 350 °C, 150 atm → phenol** (Dow process). →u11

Alcohols — classification and mechanisms

- **Hydroquinone** is the phenol with two –OH groups at **positions 1 and 4** (catechol = 1,2; resorcinol = 1,3).
- **Dehydration of an alcohol gives an alkene**; the **rate of dehydration is 3° > 2° > 1°**. ×2
- **R–C(OH)(R)–R with the –OH carbon attached to three groups = a tertiary alcohol**; **$CH_3CH(OH)CH_3$ is the secondary alcohol** in the papers' set.
- **Only a primary alcohol oxidises to an aldehyde with the same number of carbons — $CH_3CH_2CH_2OH$ → propanal**.
- **Isopropanol oxidises to acetone** (the alcohol behind a C=O with two methyls).
- **Secondary alcohols oxidise to ketones**; **carboxylic acids are reduced to primary alcohols only by $LiAlH_4$**. ×2
- **Alkaline hydrolysis of bromoethane → ethanol uses dilute aqueous NaOH, warm**; the **aqueous-KOH route is substitution and the alcoholic-KOH route is elimination**. ×2
- **Formic acid + ethanol + acid catalyst → an ester** (ethyl formate). ×2
- **Ethanol + PCl_5 releasing HCl proves the hydroxyl group**.
- **Methanol and ethanol are distinguished by the iodoform test** (ethanol is positive).
- **The group that gives a yellow precipitate of triiodomethane with alkaline iodine is the methyl ketone group CH_3–CO–R**.
- **Rectified spirit (95.6%) → absolute alcohol by drying with calcium oxide**.
- **Denaturing agents for ethanol are methanol, pyridine and acetone — lactic acid is not used**.
- **Phenol was the earliest antiseptic and disinfectant**.
- **All of C_3H_7OH, C_4H_9OH and $C_4H_8{\cdot}CH_3OH$ are alcohols** ("none of the given" is the answer).

Phenol — tests

- **Aqueous phenol decolourises bromine water with a white precipitate of 2,4,6-tribromophenol**.
- **Picric acid** is the named trinitrophenol.
- **Lucas test** remains the primary/secondary/tertiary discriminator. →u15

Bled in from other units (same buckets)

- **Osmotic pressure is not a property of a liquid**; **iodine molecules are held by instantaneous dipole–induced dipole forces**; **in anisotropic crystals molar mass does not change with direction**. →u3, →u5
- **Toluene → p-chlorotoluene uses Cl_2 with $AlCl_3$**. →u14

- **For which reaction does Kc rise with temperature: $H_2 + I_2 \rightleftharpoons 2HI$**. →u6
- **Bond energy order HCl > HBr > HI**. →u10

Bled in from other units (same buckets)

- **Enthalpy of neutralisation** = enthalpy change forming 1 mol water under standard conditions. →u8
- English items misfiled into this unit by the compilation's page order (similes, "elated" = **excited**, "queasy", "buckling", "procure", "maneuver", "jigged", conjunctions): these are outside the Physics/Chemistry/Biology scope and are recorded here only so the coverage audit closes — nothing to revise from them.

Bled in from other units (same buckets)

- **1-pentyne vs 2-pentyne** are told apart by **$AgNO_3$ + NH_4OH** (terminal alkyne gives a precipitate). →u14
- **Water moderates climate via its high specific heat capacity**. →u3
- **Sigma/pi bonding is explained by MOT and VBT but NOT by VSEPR or CFT**. →u10
- **Planck's constant = 6.6262×10^{-34} J·s**. →u14 (modern physics)
- Carbocation stability order **tertiary** > **secondary** > **primary**; **diisopropyl ketone** is the IUPAC-name item. →u13
